

CALRIG · REQUIREMENTS

CalRig requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with first-order status
0.1	2026-09-25	Amish Chadha	Initial scaffold

CalRig requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be checked by calculation at TRL 3. Test conditions follow the enhanced lab conditions in the US EPA sensor testing reports (20 °C and 40 °C; 40 % and 85 % RH) so that CalRig results can be read against a known method ([EPA FAQ](#)).

The **reference room** is an indoor lab at 22 °C and 50 % RH unless a requirement states otherwise.

Table 1. Requirements with targets and first-order status (status values are estimates from CLR-PRC-001).

ID	Requirement	Target	Verification (TRL 3 or later)	First-order status
R1	Chamber temperature range	10 to 40 °C at the sensor tray, in a room at 15 to 30 °C	Heat balance calculation; later chamber log	Not met at the low end in a 30 °C room: about 15 °C reachable (estimate). Met in a room at 25 °C or below
R2	Chamber humidity range	20 to 85 % RH at any temperature from 20 to 40 °C	Moisture balance; later chamber log	Met by estimate; 20 % RH at 20 °C depends on dryer condition
R3	Stability at a set point	±0.3 °C and ±2 % RH over a 30 min hold	Controller simulation; later log	Open until TRL 3 control model
R4	Uniformity across the six bays	±0.3 °C and ±2 % RH between bays	Airflow check; later traverse with two references	Open; depends on mixing fan layout
R5	Reference temperature uncertainty	0.2 °C or better (expanded, k = 2) from 10 to 40 °C	Uncertainty budget; ice-point check	At risk: SHT45 datasheet is ±0.1 °C typical; no calibration certificate at this budget
R6	Reference humidity uncertainty	2 % RH or better (expanded) from 20 to 85 % RH, checked at four salt fixed points	Uncertainty budget; salt fixed-point checks (Greenspan, 1977)	Met by estimate if the salt checks pass
R7	Particle test range	Zero below 2 µg/m³ PM2.5; controlled decay from 300 to 5 µg/m³ in 45 min or less	Decay calculation; later run	Met by estimate (about 30 min decay)
R8	Particle reference traceability	Transfer PM sensor collocated with a regulatory or research monitor for 30 days or more, and meeting EPA PM2.5 targets there, before use as the chamber reference	Collocation record	Not met by the rig alone. Needs a collocation site; see CLR-PRC-001
R9	CO2 check	Zero check below 50 ppm with soda lime; span 400 to 2000 ppm against a reference of ±(30 ppm + 3 %)	Datasheet (Sensirion SCD30); later run	Met by estimate
R10	Capacity	Six sensor heads per run, each up to 90 x 70 x 50 mm, with power and I2C, UART or USB leads through a sealed gland	Model check	Met in the massing model
R11	Throughput	Four-point temperature and humidity sweep plus one particle decay run in 8 h or less, unattended	Timing estimate; later log	Met by estimate (about 6 h)
R12	Cost	\$300 or less in parts	Priced BOM (bom/ bom . csv)	Not met: about \$396 (indicative). Budget change proposed, awaiting Amish
R13	Size and mass	Fits a 600 x 500 mm bench area, 400 mm high or less; 12 kg or less	Model; mass estimate	Not met on length by about 10 mm: about 610 x 470 x 350 mm, about 10 kg (estimate)

ID	Requirement	Target	Verification (TRL 3 or later)	First-order status
R14	Records	Every run writes CSV (time, set points, references, each sensor) and a per-sensor report with fitted slope, offset and humidity term, EPA metrics and the reference check dates	Software review	Open (software not written at TRL 2)
R15	Electrical safety	No mains wiring in the rig; certified external 12 V supply; fused 12 V bus; peak draw 100 W or less	Design review	Met in concept: about 70 W peak (estimate)
R16	Thermal safety	Independent hardware cut-off opens the heater circuit at 50 °C air temperature or 70 °C Peltier hot side	Design review	Met in concept
R17	Aerosol and gas safety	No toxic gases; test smoke vented through the HEPA loop before the door opens; smoke source outside the chamber	Design review	Met in concept
R18	Garage-buildable	Built with a saw or laser cutter, a drill and a soldering iron; all active parts off the shelf	Design review	Met in concept

■ Requirements not met or at risk

- **R1**, cooling to 10 °C in a warm room. A single 60 W Peltier assembly cannot hold 20 K below a 30 °C room with the estimated heat leak (CLR-PRC-001, Table 3). Options are listed in CLR-PRC-001.
- **R5**, reference temperature uncertainty. Without a calibration certificate the claim rests on the datasheet and an ice-point check.
- **R8**, particle traceability. The chamber cannot make a mass reference; it transfers one from a collocation site.
- **R12**, cost. The estimate is about \$396 against a \$300 budget.
- **R13**, footprint, by about 10 mm on length; the base plate can be trimmed at TRL 3.

■ Assumptions

- Chamber heat loss about 1.0 W/K with the insulation jacket fitted (CLR-PRC-001).
- Settling is limited by the mixing air flow (about 3 L/min through the conditioners) and by moisture held on the walls and sensors.
- Test aerosol is combustion smoke; results for other aerosols (dust, sea salt) will differ and must be stated in each report.